Phase Clan's Simon Says

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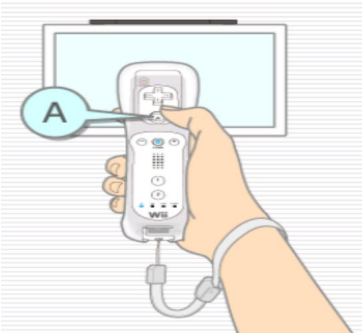
Problem

Simon Says games toys are outdated

Why Our Product?

- Portable
  - Arduino platform
  - Hook up to ANY HDMI or RCA display
- Active
  - Engaging motion controls
- Multiple modes to play
  - Play as long as you want!
  - Play against the clock!

What is Phase Clan's Simon Says game?

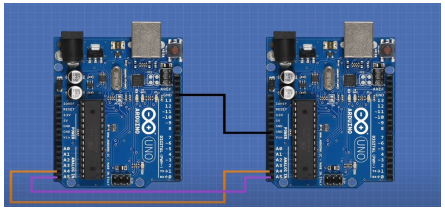


- A single player game that uses Two arduinos, a Wiimote, and display screen.
- The user tries to mimic the display screen's instructions by pressing buttons and using motions on Wiimote.

How is it implemented?

System Overview:

-The primary and secondary arduinos are connected through pins A4, A5, and GND according to I2C protocol.



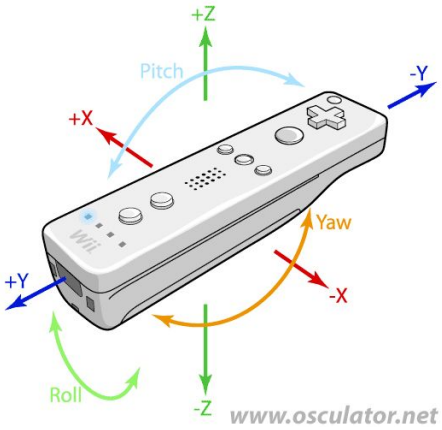
-The primary Arduino receives input from the wiimote, uses that input to run the game engine, and sends the state of the game to the secondary arduino.

-The secondary arduino receives an integer value that corresponds to which screen state to display. That state's display parameters are sent to the TV/monitor via RCA output.

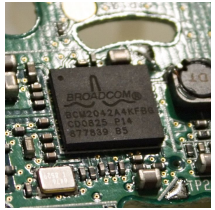
-The user presses buttons or motions according to what is on the screen

-The display shows the current state of the game to the user at a 92 x 30 resolution

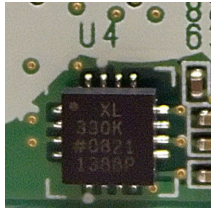
Wiimote Overview



- Bluetooth communication
- Accelerometer and Gyroscope
- Multiple buttons
- Rumble and Extensions
- Friendly form factor



Broadcom BCM2042 in a Wii remote



ADXL330 in a Wii remote

Wii Motion Plus Extension
(Gyroscope)

Excluded in RVL-003 models
Included in RVL-036 models

Arduino / Wiimote connection

- The USB host shield allows the Arduino to communicate with USB devices
- By using a USB Bluetooth dongle and USB library, communication is achievable with any bluetooth device

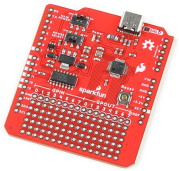
Wiimote

Inputs from accelerometer and buttons

Bluetooth Dongle

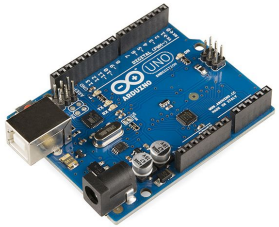
Wiimote data through USB and HID Bluetooth protocol

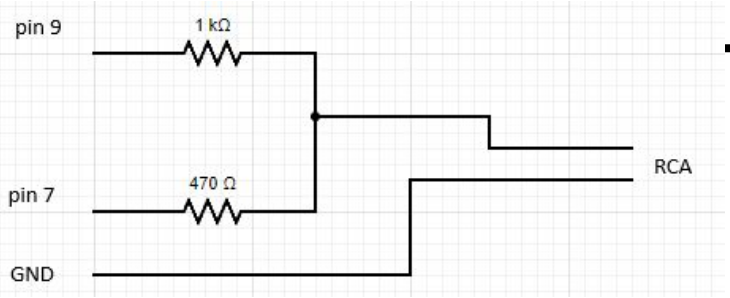
USB Host Shield

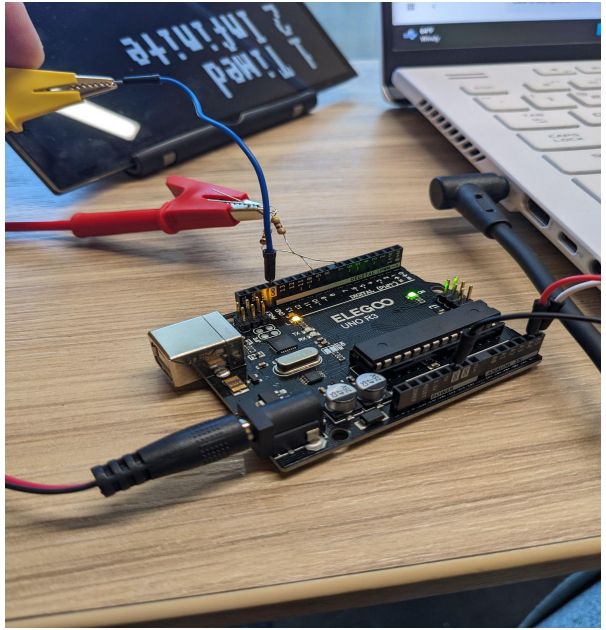


Communicate via USB Host Wii library

Arduino Uno







RCA/HDMI Overview

The secondary Arduino uses:

- 3 wire connections to primary arduino via A4, A5, and GND.
- 2 resistors: (1k, 470) ohm resistors.
  - 1k plugged into pin 9, 470 plugged into pin 7.
  - Ends twisted together to create a voltage divider
- 2 Alligator cables
  - One connected to end of resistors and RCA
  - One connected to ground and RCA
- RCA cable
  - One end dissected to expose lead wires
- Optional: RCA to HDMI signal converter

Inside the Arduinos...

Primary Arduino:

- A state machine takes input to switch between the two modes
- The two different modes cycle through random inputs for the display and player to show and replicate, respectively
- Inputs are read from smaller state machines
- 50 seconds of play time for Time Attack Mode
- Increasing difficulty for Infinite Mode

Secondary Arduino:

- Uses a state machine to control which image is being transmitted to the display
- Waits in ready state for input from primary arduino
- Once it receives input, it clears the frame buffer and transitions to a display state
- Each of the 22 display states transmits an image to the physical display.

```
void EX_Tick() {
  char buffer[20];

  switch (EX_State) {
    case preReady:
        if (count != -1){
          EX_State = transition(count);
          TV.clear_screen();
        }

      break;
```

Performance/Evaluation

- Latency:
  - Wiimote to primary arduino: 100ms
  - Primary Arduino to secondary arduino: <1ms
  - Secondary Arduino to display: <1ms
    - Total: ~100ms
- Can start a game, play it all the way through, and then continue play with no major gameplay glitches

Conclusion

This variation of the Simon Says game is a fresh alternative to the current products on the market.

Its portability and engaging play style provide easy fun for everybody.

Questions?